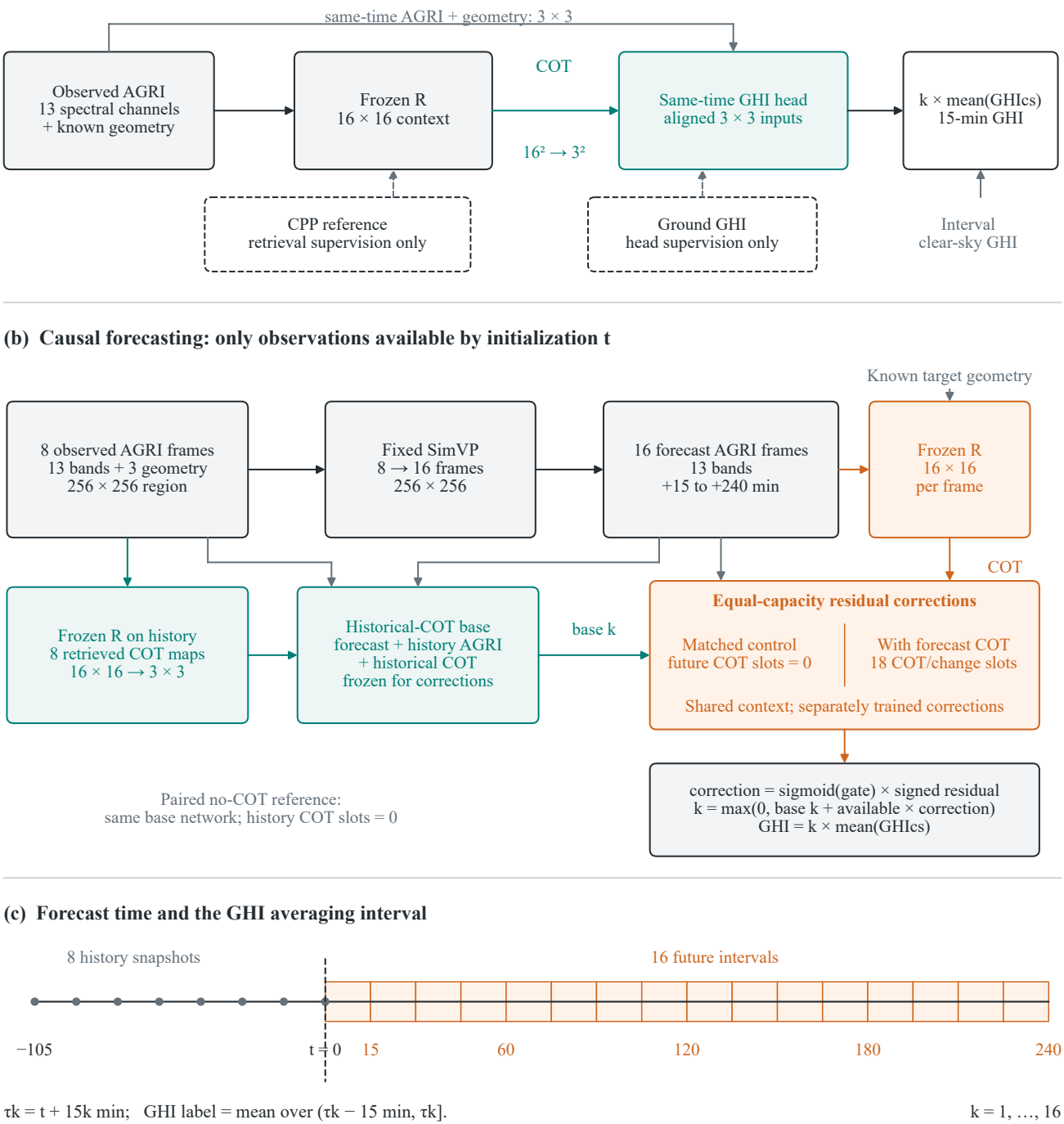

(a) Same-time COT value: observed images at target time τ
same-time AGRI + geometry: 3 × 3
Observed AGRI
13 spectral channels
+ known geometry
Frozen R
16 × 16 context
COT
16² → 3²
Same-time GHI head
aligned 3 × 3 inputs
k × mean(GHIcs)
15-min GHI
CPP reference
retrieval supervision only
Ground GHI
head supervision only
Interval
clear-sky GHI
(b) Causal forecasting: only observations available by initialization t
Known target geometry
8 observed AGRI frames
13 bands + 3 geometry
256 × 256 region
Fixed SimVP
8 → 16 frames
256 × 256
16 forecast AGRI frames
13 bands
+15 to +240 min
Frozen R
16 × 16
per frame
COT
Frozen R on history
8 retrieved COT maps
16 × 16 → 3 × 3
Historical-COT base
forecast + history AGRI
+ historical COT
frozen for corrections
base k
Equal-capacity residual corrections
Matched control
future COT slots = 0
With forecast COT
18 COT/change slots
Shared context; separately trained corrections
Paired no-COT reference:
same base network; history COT slots = 0
correction = sigmoid(gate) × signed residual
k = max(0, base k + available × correction)
GHI = k × mean(GHIcs)
(c) Forecast time and the GHI averaging interval
8 history snapshots
16 future intervals
−105
t = 0
15
60
120
180
240
τk = t + 15k min; GHI label = mean over (τk − 15 min, τk].
k = 1, …, 16